

```
# importing major libraries
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
```

```
# additional libraries
```

```
import warnings
warnings.filterwarnings('ignore')
```

```
# loading dataset
```

```
df = pd.read_csv('Magicbricks.csv')
```

```
# overview
```

```
df.head()
```

	Area	BHK	Bathroom	Furnishing	Location	District \
0	950.0	2	2.0	Furnished	Karol Bagh	Central Delhi
1	535.0	2	1.0	Furnished	Karol Bagh	Central Delhi
2	1280.0	3	3.0	Furnished	Karol Bagh	Central Delhi
3	1135.0	3	3.0	Furnished	Karol Bagh	Central Delhi
4	1135.0	3	3.0	Furnished	Karol Bagh	Central Delhi

	Status \	Locality	Parking
0	DDA MIG Flats Prasad Nagar Phase 2, Prasad Nag...		1.0
1	Ready_to_move	Dev Nagar, Karol Bagh	1.0
2	Ready_to_move	Karol Bagh	2.0
3	Almost_ready	The Amaryllis, Karol Bagh	2.0
4	Almost_ready	The Amaryllis, Karol Bagh	2.0

	Transaction	Type	Per_Sqft	Price
0	Resale	Apartment	8761.0	12500000
1	New_Property	Apartment	7290.0	3900000
2	Resale	Builder_Floor	14092.0	15000000
3	Resale	Apartment	22222.0	25000000
4	Resale	Apartment	22222.0	25000000

DAP -- Data Analysis Process

- Problem

- Data wrangling
 - Data Gathering(api,db,warehouse,csv)
 - Data Assessing
 - Data Cleaning
- Exploratory Analysis
 - Statistical Methods
 - Graphs
- Reports
- Dashboards --> Power Bi and Tableau
- Advisories

```
# data assessing
```

```
df.head()
```

	Area	BHK	Bathroom	Furnishing	Location	District \
0	950.0	2	2.0	Furnished	Karol Bagh	Central Delhi
1	535.0	2	1.0	Furnished	Karol Bagh	Central Delhi
2	1280.0	3	3.0	Furnished	Karol Bagh	Central Delhi
3	1135.0	3	3.0	Furnished	Karol Bagh	Central Delhi
4	1135.0	3	3.0	Furnished	Karol Bagh	Central Delhi

	Status \	Locality	Parking
0	DDA MIG Flats Prasad Nagar Phase 2, Prasad Nag...		1.0
1	Ready_to_move	Dev Nagar, Karol Bagh	1.0
2	Ready_to_move	Karol Bagh	2.0
3	Almost_ready	The Amaryllis, Karol Bagh	2.0
4	Almost_ready	The Amaryllis, Karol Bagh	2.0

	Transaction	Type	Per_Sqft	Price
0	Resale	Apartment	8761.0	12500000
1	New_Property	Apartment	7290.0	3900000
2	Resale	Builder_Floor	14092.0	15000000
3	Resale	Apartment	22222.0	25000000
4	Resale	Apartment	22222.0	25000000

```
# columns
```

```
df.columns
```

```
Index(['Area', 'BHK', 'Bathroom', 'Furnishing', 'Location',  
'District',  
      'Locality', 'Parking', 'Status', 'Transaction', 'Type',  
'Per_Sqft',
```

```
'Price'],  
dtype='object')
```

MagicBricks Dataset – Data Card

Dataset Name: MagicBricks Housing Dataset **Domain:** Real Estate **Purpose:** Used to analyze housing prices and build models to **predict property price** based on property features.

Columns Description

- **Area** – Size of the property (in square feet).
- **BHK** – Number of Bedrooms, Hall, and Kitchen.
- **Bathroom** – Number of bathrooms in the property.
- **Furnishing** – Furnishing status (Furnished / Semi-Furnished / Unfurnished).
- **Location** – Area where the property is located.
- **District** – District of the property.
- **Locality** – Specific neighborhood within the location.
- **Parking** – Availability of parking space.
- **Status** – Construction status (Ready to Move / Under Construction).
- **Transaction** – Type of transaction (New Property / Resale).
- **Type** – Type of property (Apartment, Builder Floor, Villa, etc.).
- **Per_Sqft** – Price per square foot.
- **Price** – Total price of the property (**Target Variable**).

```
# manual assessing  
# seeking info
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 1214 entries, 0 to 1213  
Data columns (total 13 columns):  
#   Column          Non-Null Count  Dtype  
---  -  
0   Area            1214 non-null   float64  
1   BHK             1214 non-null   int64  
2   Bathroom        1212 non-null   float64  
3   Furnishing      1214 non-null   object  
4   Location        1214 non-null   object  
5   District        1214 non-null   object  
6   Locality        1214 non-null   object  
7   Parking         1182 non-null   float64  
8   Status          1214 non-null   object  
9   Transaction     1214 non-null   object  
10  Type            1209 non-null   object  
11  Per_Sqft        973 non-null    float64
```

```
12 Price      1214 non-null  int64
dtypes: float64(4), int64(2), object(7)
memory usage: 123.4+ KB
```

Dataset Information

- The dataset consists of **1,214 rows and 13 columns**, representing the observations and features used for analysis.
- The dataset contains **null (missing) values**, which may require appropriate data cleaning techniques such as imputation or removal before performing analysis or modeling.
- A **validity issue** is observed where the **Bathroom** and **Parking** columns are stored in **float data type**, even though these values should logically be integers since they represent counts.
- The dataset includes **6 numerical columns** used for quantitative analysis and **7 object-type columns** that represent categorical or textual information.

```
# messy data vs dirty data
```

Dirty Data Issues

Dirty data refers to data that is **incorrect, incomplete, inconsistent, or unrealistic**, which can affect analysis and machine learning results.

1. Validity Issues

Validity problems occur when data **does not follow logical rules or defined formats**.

Examples:

- Duplicate entries such as the same **Patient ID appearing multiple times**
- Incorrect **data types** (e.g., age stored as text instead of numbers)
- **Negative values** where they are not possible (e.g., age = -5)

These issues make the dataset **invalid for analysis** and must be corrected.

2. Accuracy Issues

Accuracy issues occur when the value is **technically possible but not logically correct in real life**.

Examples:

- Adult weight recorded as **15 kg**
- Age recorded as **202 years**

Although these values are numbers and not technically wrong in format, they are **not realistic**, which reduces data reliability.

3. Completeness Issues

Completeness issues occur when **important data is missing or blank**.

Examples:

- Missing values in columns like **age, address, salary, or patient diagnosis**
- Empty cells in datasets

Incomplete data can lead to **biased analysis or incorrect conclusions**.

4. Inconsistency Issues

Inconsistency occurs when the **same information is represented in different formats**.

Examples:

- Name variations: **Ujjwal, Ujjawal**
- Location variations: **NYC, New York City**

These inconsistencies create problems in **grouping, filtering, and analysis**.

seeking description of the dataset

```
df.describe()
```

	Area	BHK	Bathroom	Parking
Per_Sqft \				
count	1214.000000	1214.000000	1212.000000	1182.000000
973.000000				
mean	1451.850751	2.778418	2.523927	1.708122
15574.885920				
std	1586.472855	0.946811	1.017723	5.717177
21574.389007				
min	28.000000	1.000000	1.000000	1.000000
1259.000000				
25%	800.000000	2.000000	2.000000	1.000000
6154.000000				
50%	1150.000000	3.000000	2.000000	1.000000
10838.000000				
75%	1620.000000	3.000000	3.000000	2.000000
17647.000000				
max	24300.000000	10.000000	7.000000	114.000000
183333.000000				

	Price
count	1.214000e+03
mean	2.079898e+07
std	2.561308e+07
min	1.000000e+06
25%	5.600000e+06

50%	1.400000e+07
75%	2.500000e+07
max	2.400000e+08

Potential Issues in the Dataset

- The **maximum value in the Area column is 24,300**, which appears unusually high compared to typical residential property sizes. This may indicate the presence of **outliers or possible data entry errors** that should be investigated.
- The **Parking column contains a value of 114**, which is highly unrealistic for a residential property. This suggests a **potential data quality issue or incorrect entry** that requires validation or correction before analysis.

```
# null values
```

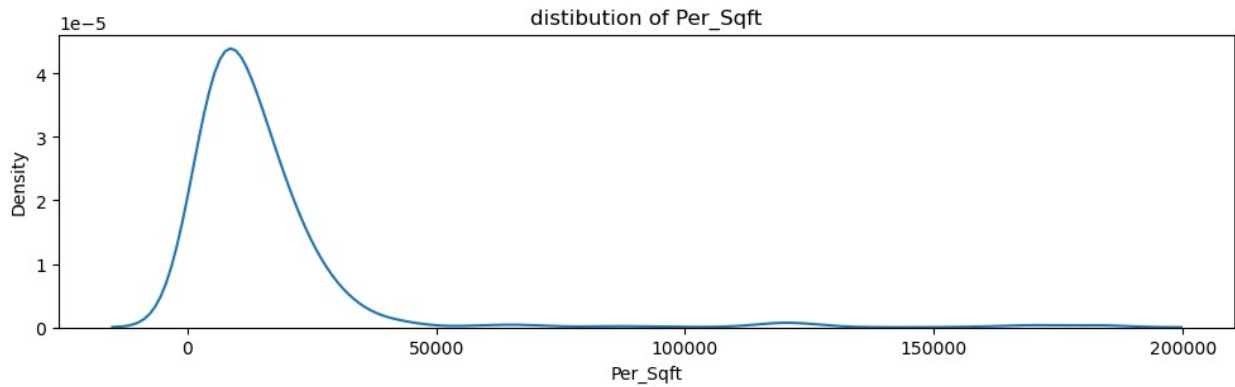
```
temp = df.isnull().sum()[df.isnull().sum()>0]  
temp
```

```
Bathroom      2  
Parking       32  
Type           5  
Per_Sqft      241  
dtype: int64
```

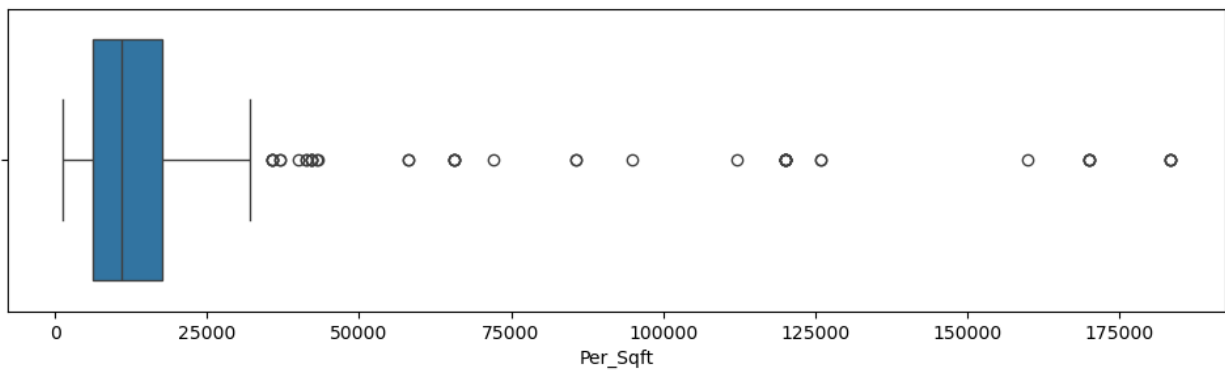
```
df.isnull().mean()*100
```

```
Area           0.000000  
BHK            0.000000  
Bathroom       0.164745  
Furnishing     0.000000  
Location       0.000000  
District       0.000000  
Locality       0.000000  
Parking        2.635914  
Status         0.000000  
Transaction    0.000000  
Type           0.411862  
Per_Sqft       19.851730  
Price          0.000000  
dtype: float64
```

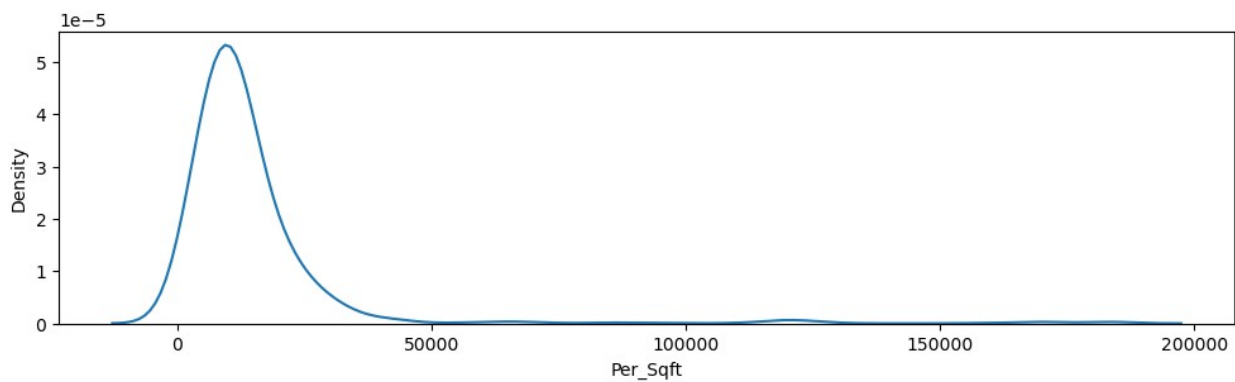
```
df.Per_Sqft  
plt.figure(figsize=(12,3))  
sns.kdeplot(data=df,x='Per_Sqft')  
plt.title('distribution of Per_Sqft')  
plt.show()  
print('skewness',df.Per_Sqft.skew().round(2))  
plt.figure(figsize=(12,3))  
sns.boxplot(data=df,x='Per_Sqft')  
plt.show()
```

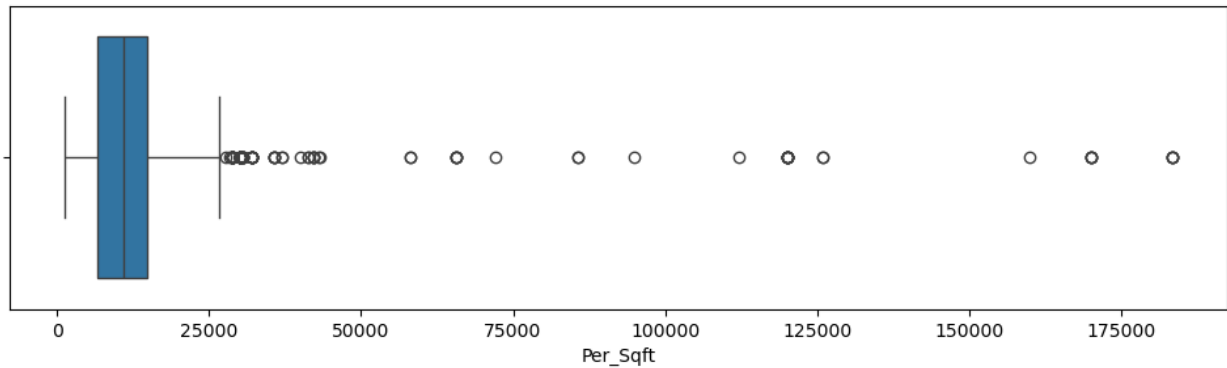


skewness 5.26



```
# not an idol way
plt.figure(figsize=(12,3))
sns.kdeplot(x=df.Per_Sqft.fillna(10838))
plt.show()
plt.figure(figsize=(12,3))
sns.boxplot(x=df.Per_Sqft.fillna(10838))
plt.show()
```





df

	Area	BHK	Bathroom	Furnishing	Location	
District \						
0	950.0	2	2.0	Furnished	Karol Bagh	Central Delhi
1	535.0	2	1.0	Furnished	Karol Bagh	Central Delhi
2	1280.0	3	3.0	Furnished	Karol Bagh	Central Delhi
3	1135.0	3	3.0	Furnished	Karol Bagh	Central Delhi
4	1135.0	3	3.0	Furnished	Karol Bagh	Central Delhi
...	...	...	...	...	...	...
1209	4688.0	5	4.0	Unfurnished	Patel Nagar	West Delhi
1210	3901.0	4	4.0	Unfurnished	Patel Nagar	West Delhi
1211	2306.0	3	3.0	Unfurnished	Patel Nagar	West Delhi
1212	675.0	2	1.0	Unfurnished	Uttam Nagar	West Delhi
1213	585.0	2	2.0	Unfurnished	Uttam Nagar	West Delhi

	Locality			Parking \
0	DDA MIG Flats Prasad Nagar Phase 2, Prasad Nag...			1.0
1	Dev Nagar, Karol Bagh			1.0
2	Karol Bagh			2.0
3	The Amaryllis, Karol Bagh			2.0
4	The Amaryllis, Karol Bagh			2.0
...	...			...
1209	The Leela Sky Villas, Patel Nagar			1.0
1210	The Leela Sky Villas, Patel Nagar			1.0
1211	The Leela Sky Villas, Patel Nagar			1.0
1212	Uttam Nagar			1.0
1213	Uttam Nagar			1.0

	Status	Transaction	Type	Per_Sqft	Price
0	Ready_to_move	Resale	Apartment	8761.0	12500000
1	Ready_to_move	New_Property	Apartment	7290.0	3900000
2	Ready_to_move	Resale	Builder_Floor	14092.0	15000000
3	Almost_ready	Resale	Apartment	22222.0	25000000
4	Almost_ready	Resale	Apartment	22222.0	25000000
...	...	...	...	...	...
1209	Almost_ready	New_Property	Apartment	28797.0	135000000
1210	Almost_ready	New_Property	Apartment	23840.0	93000000
1211	Almost_ready	New_Property	Apartment	24718.0	57000000
1212	Ready_to_move	Resale	Builder_Floor	5794.0	2900000
1213	Ready_to_move	Resale	Builder_Floor	5794.0	3200000

[1214 rows x 13 columns]

```
# Area
# 950.0 ---> 12500000
# 1 --> 12500000/950.0

# temp_l1 = df[df.Per_Sqft.isnull()].index
# df.Per_Sqft.fillna(df.price/df.area).iloc[temp_l1].round(2)
df.Per_Sqft = df.Per_Sqft.fillna(df.Price/df.Area).round(2)

df.isnull().sum()
```

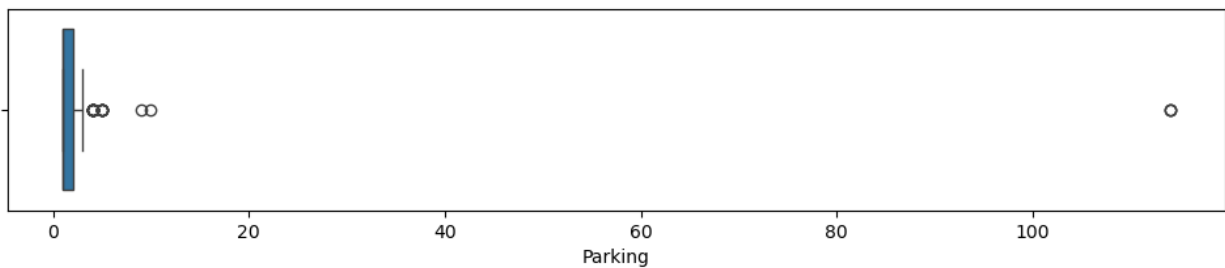
```
Area      0
BHK       0
Bathroom  2
Furnishing 0
Location  0
District  0
Locality  0
Parking   32
Status    0
Transaction 0
Type      5
Per_Sqft  0
```

```

Price          0
dtype: int64

df.Parking.describe()
plt.figure(figsize=(12,2))
sns.boxplot(x=df.Parking)
plt.show()

```

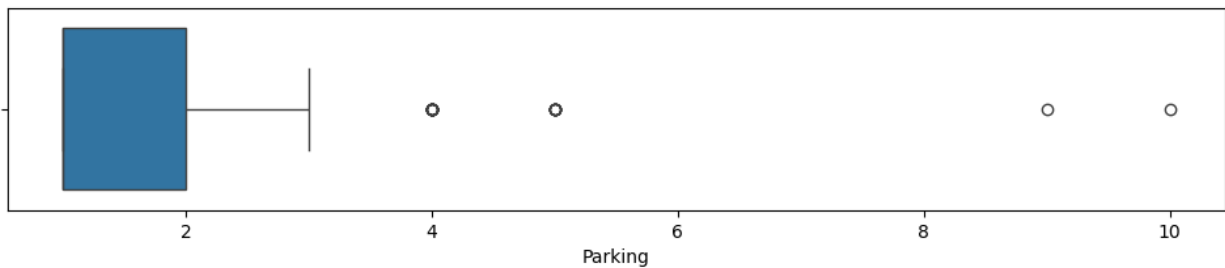


```

df[df.Parking>20]
df.Parking = np.where(df.Parking>20,1,df.Parking)

plt.figure(figsize=(12,2))
sns.boxplot(x=df.Parking)
plt.show()

```



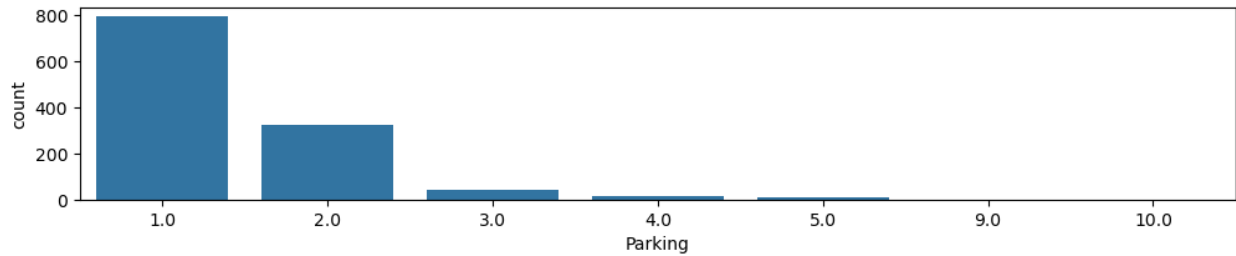
```

df.Parking.describe()

count    1182.000000
mean      1.421320
std       0.757789
min       1.000000
25%       1.000000
50%       1.000000
75%       2.000000
max       10.000000
Name: Parking, dtype: float64

df.Parking.describe()
plt.figure(figsize=(12,2))
sns.countplot(x=df.Parking)
plt.show()

```



```
df.Parking.median()
df.Parking.fillna(1,inplace=True)

df.isnull().sum()
# dropna
df.dropna(inplace=True)

df.shape
(1207, 13)

# validity issues

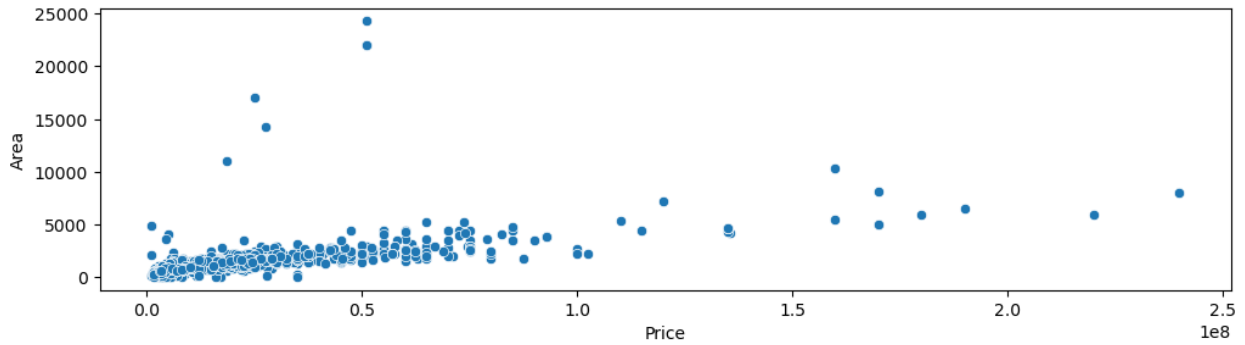
df.dtypes

df.Bathroom = df.Bathroom.astype(int)
df.Parking = df.Parking.astype(int)

df.dtypes
Area                float64
BHK                 int64
Bathroom            int64
Furnishing          object
Location            object
District            object
Locality            object
Parking             int64
Status              object
Transaction         object
Type                object
Per_Sqft            float64
Price               int64
dtype: object

# price and area comparisons

df.Price
df.Area
plt.figure(figsize=(12,3))
sns.scatterplot(data=df,x='Price',y='Area')
plt.show()
```



```
df[df.Area>10000]
```

	Area	BHK	Bathroom	Furnishing	Location	
District \						
429 Delhi	22050.0	4	4	Semi-Furnished	Greater Kailash	South
431 Delhi	22050.0	4	4	Semi-Furnished	Greater Kailash	South
515 Delhi	10350.0	4	7	Semi-Furnished	Friends Colony	South
603 Delhi	24300.0	4	5	Semi-Furnished	Saket	South
806 Delhi	14220.0	3	3	Semi-Furnished	Paschim Vihar	West
835 Delhi	17010.0	3	3	Semi-Furnished	Punjabi Bagh	West
978 Delhi	11050.0	3	3	Unfurnished	Chittaranjan Park	South

	Locality	Parking	Status
Transaction \			
429 New_Property	Greater Kailash 1	2	Almost_ready
431 New_Property	Greater Kailash 1	2	Almost_ready
515 New_Property	Maharani Bagh, New Friends Colony	3	Ready_to_move
603 Resale	Saket	2	Ready_to_move
806 New_Property	Paschim Vihar Block B4	1	Ready_to_move
835 Resale	Punjabi Bagh West	2	Ready_to_move
978 New_Property	Chittaranjan Park	1	Ready_to_move

	Type	Per_Sqft	Price
429	Builder_Floor	30556.0	51000000
431	Builder_Floor	30556.0	51000000

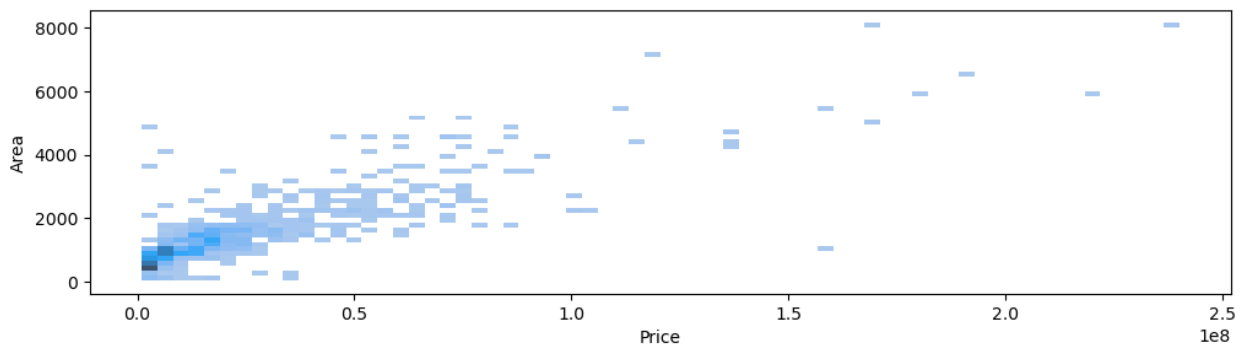
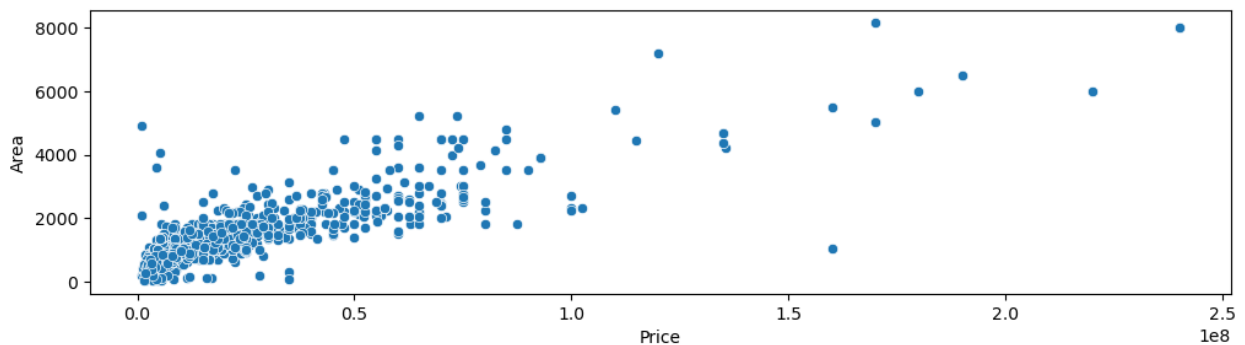
515	Apartment	15459.0	160000000
603	Builder_Floor	12500.0	51000000
806	Builder_Floor	10943.0	27500000
835	Builder_Floor	15278.0	25000000
978	Builder_Floor	12916.0	18500000

```
(22050*30556.0) / 51000000
```

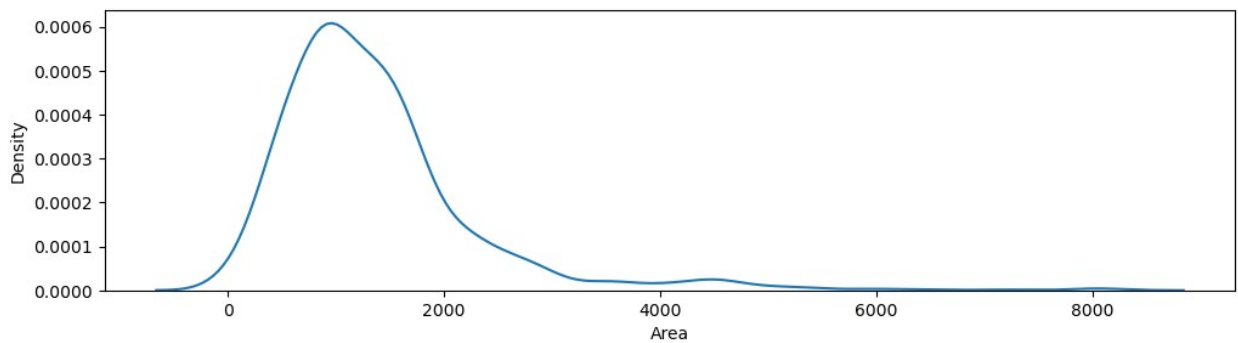
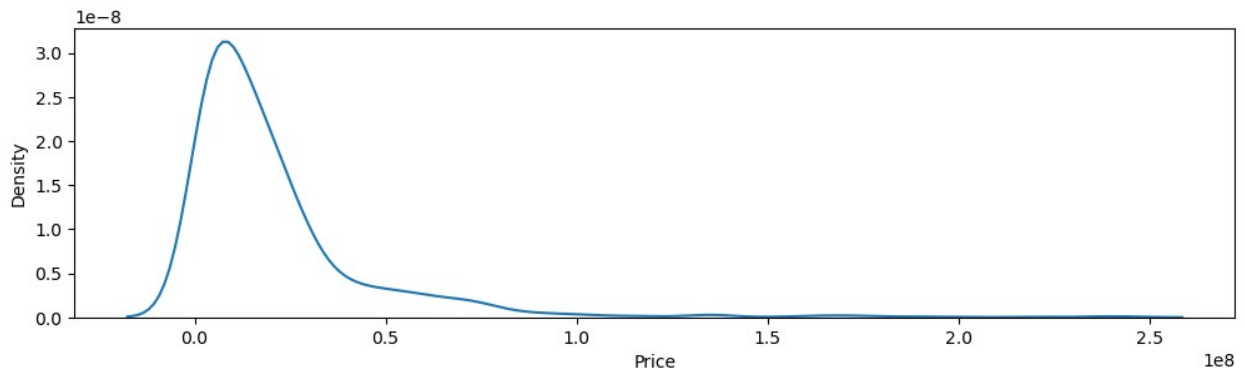
```
13.210976470588236
```

```
df.Area = np.where(df.Area>10000,df.Area/10,df.Area)
```

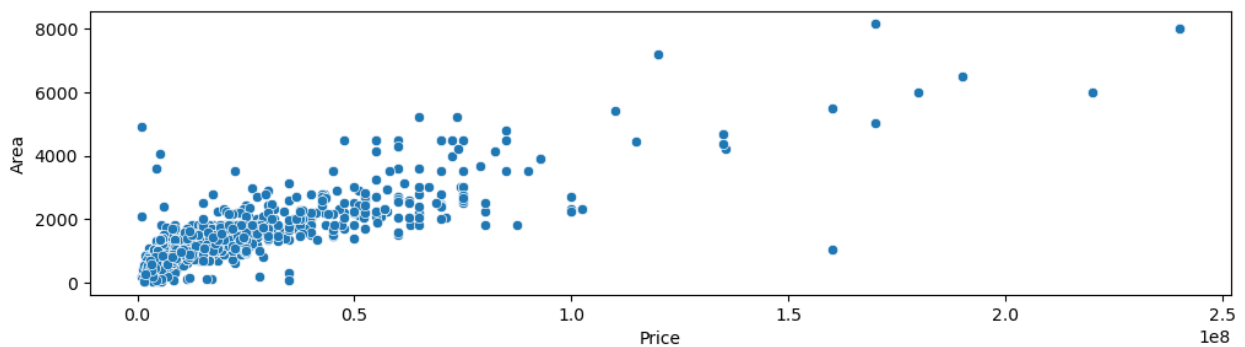
```
plt.figure(figsize=(12,3))
sns.scatterplot(data=df,x='Price',y='Area')
plt.show()
plt.figure(figsize=(12,3))
sns.histplot(data=df,x='Price',y='Area')
plt.show()
```



```
plt.figure(figsize=(12,3))
sns.kdeplot(df.Price)
plt.show()
plt.figure(figsize=(12,3))
sns.kdeplot(df.Area)
plt.show()
```



```
plt.figure(figsize=(12,3))
sns.scatterplot(data=df,x='Price',y='Area')
plt.show()
```



```
filtered_df = df[(df.Area > 2000) & (df.Price < 10000000)]
filtered_df
```

	Area	BHK	Bathroom	Furnishing	Location
District \					
37 Delhi	4050.0000	6	3	Furnished	Budh Vihar North West
39 Delhi	2400.0000	6	4	Furnished	Budh Vihar North West
387 Delhi	3600.0000	2	2	Semi-Furnished	Chhattarpur South

912	2077.4327	1	1	Unfurnished	Narela	North West Delhi
918	4900.0000	1	1	Unfurnished	Narela	North West Delhi

		Locality	Parking	\
37		Budh Vihar Phase 1	1	
39		Budh Vihar Phase 2	1	
387	Chhattarpur carpet area 400 sqyrd	status Ready...	1	
912		DDA Flats Pocket 3, Narela	1	
918		DDA Lig Flats, Narela	1	

	Status	Transaction	Type	Per_Sqft	Price
37	Ready_to_move	Resale	Builder_Floor	1259.0	5100000
39	Ready_to_move	Resale	Builder_Floor	2500.0	6000000
387	Ready_to_move	New_Property	Builder_Floor	1250.0	4500000
912	Ready_to_move	Resale	Builder_Floor	5051.0	1000000
918	Ready_to_move	New_Property	Apartment	5051.0	1000000

row 912 and 918 have problem

df.Area.describe()

```
count    1207.000000
mean     1362.493631
std       953.446919
min        28.000000
25%       800.000000
50%      1150.000000
75%      1620.000000
max       8150.000000
Name: Area, dtype: float64
```

df[df.Area<300].shape

```
df= df[df.Area>300]
df
```

	Area	BHK	Bathroom	Furnishing	Location
District \					
0	950.0	2	2	Furnished	Karol Bagh Central Delhi
1	535.0	2	1	Furnished	Karol Bagh Central Delhi
2	1280.0	3	3	Furnished	Karol Bagh Central Delhi
3	1135.0	3	3	Furnished	Karol Bagh Central Delhi
4	1135.0	3	3	Furnished	Karol Bagh Central Delhi
...	...	...	...	...	...

1209	4688.0	5	4	Unfurnished	Patel Nagar	West Delhi
1210	3901.0	4	4	Unfurnished	Patel Nagar	West Delhi
1211	2306.0	3	3	Unfurnished	Patel Nagar	West Delhi
1212	675.0	2	1	Unfurnished	Uttam Nagar	West Delhi
1213	585.0	2	2	Unfurnished	Uttam Nagar	West Delhi

		Locality	Parking \
0	DDA MIG Flats Prasad Nagar Phase 2,	Prasad Nag...	1
1		Dev Nagar, Karol Bagh	1
2		Karol Bagh	2
3		The Amaryllis, Karol Bagh	2
4		The Amaryllis, Karol Bagh	2
...		...	...
1209		The Leela Sky Villas, Patel Nagar	1
1210		The Leela Sky Villas, Patel Nagar	1
1211		The Leela Sky Villas, Patel Nagar	1
1212		Uttam Nagar	1
1213		Uttam Nagar	1

	Status	Transaction	Type	Per_Sqft	Price
0	Ready_to_move	Resale	Apartment	8761.0	12500000
1	Ready_to_move	New_Property	Apartment	7290.0	3900000
2	Ready_to_move	Resale	Builder_Floor	14092.0	15000000
3	Almost_ready	Resale	Apartment	22222.0	25000000
4	Almost_ready	Resale	Apartment	22222.0	25000000
...	...	...	...	...	...
1209	Almost_ready	New_Property	Apartment	28797.0	135000000
1210	Almost_ready	New_Property	Apartment	23840.0	93000000
1211	Almost_ready	New_Property	Apartment	24718.0	57000000
1212	Ready_to_move	Resale	Builder_Floor	5794.0	2900000
1213	Ready_to_move	Resale	Builder_Floor	5794.0	3200000

[1177 rows x 13 columns]


```
df.describe()
```

```
df[df.Per_Sqft>=183333]
```

	Area	BHK	Bathroom	Furnishing	Location	District
507	2700.0	4	4	Semi-Furnished	Lajpat Nagar	South Delhi
508	1400.0	3	3	Semi-Furnished	Lajpat Nagar	South Delhi
1028	1400.0	3	3	Unfurnished	Lajpat Nagar	South Delhi
1029	1800.0	3	3	Unfurnished	Lajpat Nagar	South Delhi

Type \	Locality	Parking	Status	Transaction	
507	Lajpat Nagar	3	2	Ready_to_move	Resale
508	Lajpat Nagar	3	1	Ready_to_move	New_Property
1028	Lajpat Nagar	3	1	Ready_to_move	Resale
1029	Lajpat Nagar	3	1	Ready_to_move	Resale

Apartment

	Per_Sqft	Price
507	183333.0	55000000
508	183333.0	32500000
1028	183333.0	24500000
1029	183333.0	24500000

```
df.shape
```

```
(1177, 13)
```

```
df.District.value_counts()
```

```
District
South Delhi      454
West Delhi       144
East Delhi       129
North West Delhi 112
South West Delhi  93
South West Delhi  87
Shahdara         68
South East Delhi  58
Central Delhi    32
Name: count, dtype: int64
```

```
df.District.unique()      # Incosistency
```

```
array(['Central Delhi', 'East Delhi', 'North West Delhi', 'Shahdara',  
      'South Delhi', 'South East Delhi', 'South West Delhi',  
      'South West Delhi ', 'West Delhi'], dtype=object)
```

```
df.District.unique()  
len(df.District.unique())
```

```
9
```

```
df.District = df.District.str.strip()  
len(df.District.unique())
```

```
8
```

```
# clean data
```

```
df.to_csv('magic_bricks_clean.csv', index=False)
```